BAB-GJR

Conditional Betting Against Beta

Evaluating GJR-GARCH Conditional Betas for the BAB Factor

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Course Project

Period: January 2000 – December 2022

In-Sample: 2000–2015 | Out-of-Sample: 2016–2022

Table of Contents

1. Executive Summary

This report evaluates whether replacing conventional rolling OLS betas with conditional betas derived from a GJR-GARCH(1,1) market volatility model and EWMA stock-market covariances improves the performance of the Betting Against Beta (BAB) factor of Frazzini and Pedersen (2014). The analysis covers CRSP common shares with market capitalisation above USD 200 million over the period January 2000 to December 2022, split into an in-sample estimation window (2000–2015) and an out-of-sample evaluation window (2016–2022).

The conditional BAB-GJR strategy delivers an annualised return of 9.05% with a Sharpe ratio of 0.69 over the full sample, compared with 4.78% and 0.55 for BAB-OLS. The improvement is most pronounced out of sample, where BAB-GJR earns 9.20% per annum (Sharpe 0.54) versus only 1.05% (Sharpe 0.12) for the OLS benchmark. However, these gains come with materially higher turnover (annualised 1,868% versus 278%) and elevated higher moments (kurtosis of 47.69 versus 1.45). The Jobson–Korkie test for the difference in out-of-sample Sharpe ratios yields a z-statistic of 0.838 ($p = 0.402$), so the improvement is economically meaningful but not statistically significant at conventional levels.

2. Introduction and Motivation

The low-beta anomaly—the empirical observation that low-beta stocks earn higher risk-adjusted returns than high-beta stocks—is one of the most robust findings in asset pricing. Frazzini and Pedersen (2014) formalised this insight into the Betting Against Beta (BAB) factor, which goes long leveraged low-beta stocks and short deleveraged high-beta stocks to achieve a market-neutral, self-financing portfolio.

The standard implementation estimates betas via rolling OLS-style computations: a one-year window for individual volatilities and a five-year window for correlations with three-day overlapping returns. While this approach is robust and widely adopted, it is inherently backward-looking and slow to adapt to regime changes such as the Global Financial Crisis (GFC) or the COVID-19 crash.

This project investigates whether replacing the static OLS beta with a conditional beta improves BAB performance. The conditional beta combines two components: a GJR-GARCH(1,1) model for market variance (capturing the well-documented leverage effect in aggregate volatility) and an exponentially weighted moving average (EWMA) for stock-market covariance (providing faster adaptation to time-varying co-movement). The GJR-GARCH parameters are estimated once on the in-sample period and then frozen, ensuring a strict separation between model calibration and out-of-sample evaluation.

3. Data and Universe

3.1 Asset Universe

The investment universe consists of CRSP US common shares (share codes 10 and 11) listed on NYSE, AMEX, or NASDAQ (exchange codes 1, 2, 3). A market capitalisation filter of USD 200 million is applied using previous month-end market cap, with eligibility shifted forward one month to avoid look-ahead bias. No S&P 500 constituent filter is used.

3.2 Data Sources

Daily stock returns, prices, and shares outstanding are sourced from WRDS/CRSP. The market return is the CRSP value-weighted index return from the daily index file. The risk-free rate and Fama–French three-factor data (Mkt–RF, SMB, HML) are obtained from the Fama–French monthly factors dataset.

3.3 Data Preparation

Daily simple returns are converted to log-returns via $r = \ln(1 + R)$. Monthly stock returns are constructed by compounding daily returns within each calendar month. Excess returns are computed only at the portfolio construction stage by subtracting the monthly risk-free rate from each portfolio sleeve return.

4. Methodology

4.1 Benchmark: BAB-OLS (Frazzini & Pedersen)

4.1.1 Beta Estimation

At each month-end, OLS betas are estimated using the decomposition $\beta = \rho \times (\sigma_i / \sigma_m)$, where σ_i and σ_m are rolling one-year (252-day, minimum 120 observations) standard deviations and ρ is the rolling five-year (1,260-day, minimum 750 observations) correlation computed on three-day overlapping log-returns. Raw betas are shrunk toward one with a fixed Vasicek-style intensity:

$$\beta_i = 0.6 \times \beta_i^{raw} + 0.4$$

4.1.2 Portfolio Construction

Stocks are ranked by their shrunk beta each month. Rank scores are centred at the median, and stocks are continuously assigned to long (low-beta) and short (high-beta) sleeves with weights proportional to the absolute deviation from the median. Weights are normalised within each sleeve. The BAB return is then:

$$r\_BAB = (1/\beta\_L) \times (r\_L - r\_f) - (1/\beta\_H) \times (r\_H - r\_f)$$

This construction ensures the portfolio is approximately dollar-neutral and beta-neutral by design.

4.2 Conditional Estimator: BAB-GJR

4.2.1 GJR-GARCH(1,1) Market Volatility

A GJR-GARCH(1,1) model with Student-t innovations is estimated on daily market log-returns (scaled by 100 for numerical stability) over the in-sample period only. The conditional variance equation is:

$$\sigma^2\_m,t = \omega + \alpha\varepsilon^2\_\{m,t-1\} + \gamma\varepsilon^2\_\{m,t-1\}\cdot 1\_\{\varepsilon<0\} + \beta\sigma^2\_\{m,t-1\}$$

The estimated parameters are frozen after in-sample calibration and used to filter conditional market variance day-by-day over the full sample (both IS and OOS). Initialisation uses the unconditional variance $\sigma^2_0 = \omega / (1 - \alpha - \gamma/2 - \beta)$.

4.2.2 Estimated Parameters

Parameter	Estimate	Std. Error	t-stat	Interpretation
μ (mean)	0.0249	0.0135	1.855	Daily mean return
ω (intercept)	0.0163	–	–	Long-run variance floor
α (ARCH)	0.0000	–	–	Symmetric shock effect

γ (leverage)	0.1744	–	sig.	Asymmetric/negative shock
β (GARCH)	0.8990	–	sig.	Volatility persistence
Persistence	0.9862	–	–	$\alpha + \gamma/2 + \beta < 1$

The key finding is that $\alpha = 0$ and $\gamma = 0.174$, indicating that the ARCH effect operates exclusively through the leverage channel: only negative market shocks increase conditional variance. Persistence of 0.986 confirms that volatility shocks are highly persistent but mean-reverting.

4.2.3 EWMA Covariance

For each stock, the daily covariance with the market is computed recursively using an EWMA filter with decay factor $\lambda = 0.94$. The stock residual is proxied by the daily log-return; the market residual is the daily log-return minus the GJR mean μ_m. The recursion is initialised with the sample covariance over the first 125 trading days:

$$Cov_{i,t} = \lambda \times Cov_{i,t-1} + (1-\lambda) \times \varepsilon_{i,t} \times \varepsilon_{m,t}$$

4.2.4 Conditional Beta

At each month-end, the conditional beta is the ratio of the EWMA covariance to the GJR conditional market variance. Individual betas are clipped to [0.1, 3.0] and then shrunk toward one with the same 0.6/0.4 intensity as OLS. The BAB-GJR portfolio is constructed identically to BAB-OLS but using these conditional betas.

An additional leverage cap of 1.5 is applied at the portfolio level: sleeve betas are floored at 1/1.5 = 0.667, preventing extreme implicit leverage in stressed months.

4.3 Transaction Costs

A proportional transaction cost of 5 basis points per unit of turnover is applied monthly. Costs are computed from the sum of absolute weight changes in both the long and short sleeves. Net BAB returns are reported as gross returns minus these costs.

5. Results

5.1 Performance Summary

Strategy | Period	Ann. Ret	Ann. Vol	Sharpe	Max DD	Skew	Kurt	CVaR 5%
BAB-OLS | Full	4.78%	8.70%	0.55	-26.24%	-0.54	1.45	-5.66%
BAB-GJR | Full	9.05%	13.15%	0.69	-15.02%	5.08	47.69	-5.56%
BAB-OLS | IS	6.79%	8.50%	0.80	-16.79%	-0.42	1.40	-5.05%
BAB-GJR | IS	8.96%	10.66%	0.84	-15.02%	3.89	39.79	-5.13%
BAB-OLS | OOS	1.05%	9.00%	0.12	-26.24%	-0.73	1.51	-6.20%
BAB-GJR | OOS	9.20%	16.90%	0.54	-14.55%	5.15	39.84	-5.43%

BAB-GJR outperforms BAB-OLS on return and Sharpe ratio across all three evaluation windows. The improvement is starkest out of sample: BAB-GJR earns 9.20% per annum (Sharpe 0.54) versus 1.05% (Sharpe 0.12) for OLS. Maximum drawdown is also substantially reduced: −14.55% for GJR versus −26.24% for OLS in the OOS period.

However, BAB-GJR exhibits markedly higher kurtosis (47.69 full sample versus 1.45) and positive skewness (5.08 versus −0.54), indicating that its return distribution has fat tails driven by large positive outliers. The CVaR at the 5% level is comparable between the two strategies, suggesting similar downside tail risk.

5.2 Risk-Adjusted Returns: CAPM and FF3 Alphas

Model	BAB-OLS IS	BAB-GJR IS	BAB-OLS OOS	BAB-GJR OOS
CAPM Alpha	0.0055 (2.50)	0.0073 (3.36)	0.0012 (0.52)	0.0096 (1.40)
FF3 Alpha	0.0052 (2.39)	0.0070 (3.04)	0.0006 (0.23)	0.0092 (1.54)

BAB-GJR produces higher CAPM and FF3 alphas than BAB-OLS in both sub-periods. In-sample, the GJR CAPM alpha is 73 basis points per month (t = 3.36) and the FF3 alpha is 70 basis points (t = 3.04), both highly significant. Out of sample, the GJR alphas remain economically large (96 and 92 basis points per month, respectively) but the t-statistics fall below conventional significance thresholds (1.40 and 1.54), reflecting the shorter OOS window (84 months) and higher return volatility.

By contrast, BAB-OLS alphas are small and insignificant out of sample (CAPM alpha of 12 bps with t = 0.52; FF3 alpha of 6 bps with t = 0.23), suggesting that the unconditional beta approach struggles to generate abnormal returns in the post-2015 environment.

5.3 Gross versus Net Performance

Strategy | Period | Type	Ann. Return	Ann. Vol	Sharpe

OLS | Full | Gross	4.78%	8.70%	0.55
OLS | Full | Net	4.64%	8.70%	0.53
GJR | Full | Gross	11.55%	14.60%	0.79
GJR | Full | Net	10.69%	14.62%	0.73
OLS | OOS | Gross	1.05%	9.00%	0.12
OLS | OOS | Net	0.89%	9.01%	0.10
GJR | OOS | Gross	9.20%	16.90%	0.54
GJR | OOS | Net	8.32%	16.93%	0.49

Transaction costs erode BAB-GJR returns by approximately 86 basis points annually in the full sample, compared with only 14 basis points for OLS. This reflects the substantially higher turnover of the conditional strategy. Despite this, net-of-cost Sharpe ratios for BAB-GJR remain well above those of OLS across all periods.

5.4 Leverage and Turnover

Metric	OLS Mean	OLS Max	GJR Mean	GJR Max
Long Leverage (1/β_L)	1.208	1.614	1.306	1.500
Short Leverage (1/β_H)	0.712	0.906	0.702	1.500
Monthly Turnover	23.2%	–	155.7%	–
Annualized Turnover	278%	–	1868%	–
Avg Monthly TC	0.01%	–	0.07%	–

BAB-GJR has substantially higher turnover due to the faster-reacting EWMA covariance estimates, which cause more frequent reshuffling of beta ranks. Mean monthly turnover is 155.7% for GJR versus 23.2% for OLS. Leverage is contained by the cap at 1.5 for the GJR strategy; without capping, extreme leverage would arise in months with very low aggregate sleeve betas.

5.5 Statistical Significance: Jobson–Korkie Test

The Jobson–Korkie test for equality of Sharpe ratios in the OOS period yields the following results:

SR BAB-GJR (ann)	SR BAB-OLS (ann)	z-stat	p-value
0.545	0.116	0.838	0.402

The annualised Sharpe ratio difference of 0.43 (0.545 versus 0.116) is economically meaningful but the test fails to reject the null of equal Sharpe ratios at the 5% level (p = 0.402). This is not unexpected given the relatively short OOS window of 84 months and the high variance of Sharpe ratio estimators.

5.6 Beta Diagnostics

The cross-sectional correlation between OLS and conditional betas is 0.306, confirming that the two estimators capture meaningfully different dimensions of systematic risk. Conditional betas have a tighter distribution (standard deviation of 0.44 versus 0.31 for OLS after shrinkage, with the conditional range bounded by the [0.46, 2.20] interval from clipping) and respond more rapidly to market regime changes.

6. Discussion

The results suggest that conditional beta estimation offers a meaningful improvement to the BAB factor, particularly out of sample. The GJR-GARCH model captures the asymmetric volatility response to negative market shocks, while the EWMA covariance provides timely updates to stock-market co-movement. Together, they allow the BAB-GJR strategy to re-rank stocks more effectively during and after market stress episodes.

The main trade-off is turnover. The EWMA decay factor of 0.94 implies an effective half-life of approximately 11 trading days, which is much shorter than the rolling windows used in OLS (252 and 1,260 days). This causes frequent rebalancing and, even with a modest 5 bps cost assumption, erodes returns by nearly 90 basis points per year. In practice, implementation costs could be higher if market-impact effects are considered.

The high kurtosis of BAB-GJR returns warrants attention. The positive skewness and fat right tail suggest that a small number of months contribute disproportionately to cumulative performance. While this is preferable to negative tail risk, it implies that the strategy's realised Sharpe ratio may be sensitive to the inclusion or exclusion of a few extreme months.

Several avenues for further research remain outside the scope of this implementation: a volatility-targeting overlay to stabilise portfolio-level risk, a multi-λ EWMA robustness grid to assess sensitivity to the decay parameter, quarterly rebalancing to reduce turnover, and bootstrap inference to address the short OOS window.

7. Conclusion

This report demonstrates that conditional betas constructed from a GJR-GARCH(1,1) market volatility model and EWMA stock-market covariances substantially improve the Betting Against Beta factor relative to the standard OLS benchmark. The BAB-GJR strategy delivers higher returns, higher Sharpe ratios, and lower maximum drawdowns across all evaluation periods, with the most striking improvement occurring out of sample (annualised return of 9.20% versus 1.05%; Sharpe of 0.54 versus 0.12).

These gains come at the cost of materially higher turnover and transaction costs, and the Sharpe ratio improvement is not statistically significant at the 5% level under the Jobson–Korkie test. Nonetheless, the economic magnitude of the improvement—particularly the near-complete elimination of the BAB factor's well-documented post-2015 deterioration—provides compelling motivation for adopting conditional beta estimation in factor construction.

8. References

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